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United States Patent	12402830
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Martin; Parker Ray

Allergy testing device

Abstract

An allergy testing device is disclosed. The allergy testing device includes a first applicator, a second applicator, a capture device, and a processor. The first applicator is arranged to apply an allergen to a first predetermined area of a patient's skin. The second applicator is arranged to apply antigen epitopes within a second predetermined area of a patient's skin. The capture device has a projector and one or more cameras. The processor is in communication with the capture device and a drive device that is arranged to move the first applicator to apply the allergen to the patient's skin and to move the second applicator to apply antigen epitopes to the patient's skin.

Inventors:	Martin; Parker Ray (Ann Arbor, MI)
Applicant:	Martin; Parker Ray (Ann Arbor, MI)
Family ID:	78005496
Appl. No.:	18/640908
Filed:	April 19, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240260888 A1	Aug. 08, 2024

Related U.S. Application Data

continuation parent-doc US 17214847 20210327 US 12076156 child-doc US 18640908
us-provisional-application US 63007363 20200408

Publication Classification

Int. Cl.: A61B5/00 (20060101)

U.S. Cl.:

CPC **A61B5/411** (20130101); **A61B5/004** (20130101); **A61B5/445** (20130101); **A61B5/742** (20130101); A61B2560/0406 (20130101)

Field of Classification Search

CPC: A61B (5/411); A61B (5/004); A61B (5/445); A61B (5/742); A61B (2560/0406); A61B (5/0077)

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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Primary Examiner: Abouelela; May A

Assistant Examiner: Warsi; Yasmeeen S

Attorney, Agent or Firm: Varnum LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/214,847, filed on Mar. 27, 2021, and entitled Allergy Testing Device, which claims the benefit U.S. Provisional Application Ser. No. 63/007,363, filed Apr. 8, 2020. Each of the foregoing are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to an allergy testing device.

BACKGROUND

(2) Current testing for IgE mediated allergic reactions is a time intensive prospect requiring trained professionals for accurate administration and interpretation of test results. Frequently, patients have allergic reactions to a number of allergens including, but not limited to, food, environmental, or drug allergens which can trigger unpleasant or dangerous reactions upon exposure to the patient's immune system. The prevalence of some allergies appears to be increasing while others appear to be over diagnosed, leading to a challenging dilemma requiring clinicians to weigh caution against the negative effects of allergen avoidance.

(3) Traditional allergy testing requires lengthy visits to a specialist's office where a test can be applied and interpreted by a trained professional. This is a time-consuming process which is prone to human error. Access to allergy testing is limited in many settings and is often not available in a timely fashion despite increased patient morbidity and mortality associated with unknown or misdiagnosed allergies. Presently there is no offering of a reliable, automated administration and interpretation of allergy testing.

(4) Therefore, it would be desirable to provide a system and method that cure one or more shortfalls of the previous approaches identified above.

SUMMARY

- (5) An allergy testing device is disclosed, in accordance with one or more embodiments of the present disclosure. In one embodiment, the allergy testing device includes a first applicator, a drive device, a capture device, and a processor. The first applicator is connected to a housing. The drive device is connected to the housing and is arranged to move the first applicator between a first applicator first position and a first applicator second position. The capture device is connected to the housing and is spaced apart from the first applicator. The processor is programmed to operate the capture device to project a first predetermined set/series of patterns onto the patient's skin within the first and/or second predetermined area; operate the capture device to capture a first set/series of images of a first and/or second predetermined area of a patient's skin; and operate the drive device to move the first applicator from the first applicator first position towards the first applicator second position, within the first predetermined area.
- (6) An allergy testing device is disclosed, in accordance with one or more embodiments of the present disclosure. In one embodiment, the allergy testing device includes a first applicator, a second applicator, a capture device, and a processor. The first applicator is arranged to apply an allergen to a first predetermined area of a patient's skin. The second applicator is arranged to apply antigen epitopes within a second predetermined area of a patient's skin. The capture device has a projector and one or more cameras. The processor is in communication with the capture device and a drive device that is arranged to move the first applicator to apply the allergen to the patient's skin and to move the second applicator to apply antigen epitopes to the patient's skin.
- (7) A method of allergy testing is disclosed, in accordance with one or more embodiments of the present disclosure. In one embodiment, the method includes capturing a first image set/series of a first and/or second predetermined area of a patient's skin and applying an allergen within the first predetermined area of the patient's skin. After a first predetermined time period, the method further includes capturing a second image set/series of the first predetermined area of the patient's skin and determining a significance of the second image set/series. Responsive to the determination derived from the second image set/series being considered significant, the method further includes outputting for display a first indicator indicative of an allergic reaction.
- (8) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not necessarily restrictive of the present disclosure as claimed. The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the present disclosure and together with the general description, serve to explain the principles of the present disclosure.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The numerous advantages of the disclosure may be better understood by those skilled in the art by reference to the accompanying figures.
- (2) FIG. 1 is a simplified schematic view of an allergy testing device, in accordance with one or more embodiments of the present disclosure.
- (3) FIG. 2 is a view of first and second predetermined areas on a patient's skin.
- (4) FIG. 3 is a view of first and second predetermined patterns projected onto the first and second predetermined areas.
- (5) FIG. 4 is a simplified schematic view of the allergy testing device in a first position or retracted position.
- (6) FIG. 5 is a simplified schematic view of the allergy testing device in a second position or extended position.
- (7) FIG. 6 illustrates a flow diagram depicting a method of allergy testing, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(8) The present disclosure has been particularly shown and described with respect to certain embodiments and specific features thereof. The embodiments set forth herein are taken to be illustrative rather than limiting. It should be readily apparent to those of ordinary skill in the art that various changes and modifications in form and detail may be made without departing from the spirit and scope of the disclosure. Reference will now be made in detail to the subject matter disclosed, which is illustrated in the accompanying drawings.

(9) Referring generally to FIGS. 1, 4, and 5, an allergy testing device **100** is illustrated in accordance with one or more embodiments of the present disclosure. The allergy testing device **100** is arranged to both apply an allergy test to a patient's skin **102** as well as interpret and report the results of this test. The allergy testing device **100** includes a housing **104**, a first applicator **106**, a second applicator **108**, a drive device **110**, a capture device **112**, and at least one processor **114**.

(10) The housing **104** may include a first housing member **200** and a second housing member **202**. The first housing member **200** and the second housing member **202** may be joined together such that they are of unibody construction. The first housing member **200** and the second housing member **202** may be joined to each other such that the second housing member **202** is removable or separable from the first housing member **200**. In at least one embodiment, at least a portion of the first housing member **200** may be movable relative to the second housing member **202** due to the operation of the drive device **110**.

(11) The first housing member **200** defines a wall **210**. The wall **210** extends between a portion of the second housing member **202** and a securing device **212** that engages the patient's skin **102**. The wall **210** is disposed about and receives portions of the first applicator **106**, the second applicator **108**, and the capture device **112**. The first housing member **200** may be referred to as a lower unit or reservoir unit.

(12) The wall **210** serves multiple purposes including spacing the capture device **112** a predetermined distance from the patient's skin **102** and inhibits ambient or external light intrusion into a chamber or space defined by the inner surface of the wall **210** and the second housing member **202**. As such, the wall **210** of the first housing member **200** may be constructed of an opaque or non-transparent material.

(13) The securing device **212** may be disposed at a distal end of the wall **210** of the first housing member **200**. The securing device **212** may be an adhesive pad that engages the patient skin **102** to resist or inhibit movement of the allergy testing device **100** relative to the patient's skin **102**. The securing device **212** may be removable from the wall **210** such that it may be discarded after the allergy testing device **100** has been disconnected from the patient's skin **102** at the conclusion of an allergy test.

(14) In at least one embodiment, the securing device **212** may be operatively connected to at least one of the first housing member **200** and the second housing member **202**. In such an embodiment, the securing device **212** may be a strap, band, or the like that extends about or extends from at least one of the first housing member **200** and/or the second housing member **202** and extends about a portion of a body part of the patient, such as an arm. In such an embodiment, a distal end of the wall **210** that is disposed opposite the second housing member **202** may engage or be disposed proximate the patient's skin **102**.

(15) The second housing member **202** is spaced apart from the patient's skin **102** by the wall **210** of the first housing member **200**. The second housing member **202** is disposed about or is arranged to receive the drive device **110**, the capture device **112**, and the processor **114**. The second housing member **202** may be referred to as a technology unit.

(16) The second housing member **202** includes a protrusion **220** that is arranged to extend towards the patient's skin **102**. The protrusion **220** is disposed between the first applicator **106** and the second applicator **108**. The wall **210** is disposed about the protrusion **220**.

(17) The protrusion **220** defines a shelf **222** that is disposed generally parallel to the patient's skin

102 and is disposed generally perpendicular to the wall **210** of the first housing member **200**. The wall **210** of the first housing member **200** is arranged to space the shelf **222** a predetermined distance from the patient's skin **102**. The shelf **222** may be the lowest point of the second housing member **202**.

(18) The first applicator **106** and the second applicator **108** are at least partially disposed within the first housing member **200**. The first applicator **106** and the second applicator **108** are operatively connected to at least one of the first housing member **200** or the second housing member **202** of the housing **104**. In the embodiment shown, the first applicator **106** and the second applicator **108** are connected to the second housing member **202** and are arranged to move relative to the first housing member **200**.

(19) In some embodiments, the first applicator **106** and the second applicator **108** may be disposed proximate the distal end of the wall **210** of the first housing member **200**. In such an embodiment, a portion of the first applicator **106** and the second applicator **108** may extend beyond the distal end of the wall **210** of the first housing member **200** such that operation of the drive device **110** may move the first housing member **200** or the first applicator **106** and the second applicator **108** relative to the second housing member **202** to apply allergens or the like to a patient's skin **102**.

(20) The first applicator **106** is arranged to apply an allergen or the like to a first predetermined area **300** of the patient's skin **102**, as shown in FIGS. 2, 3, and 5. A footprint of or a cross sectional area of the first applicator **106** may be equal to or less than the first predetermined area **300**.

(21) The first applicator **106** includes a reservoir **302** and a prong or a needle **304**. The reservoir **302** contains allergens, histamines, antigen epitopes, saline (e.g. for negative control) or other that will be applied to the patient skin **102** by the at least one prong or needle **304**. The at least one prong or needle **304** may be disposed within the reservoir such that the contents of the reservoir may be soaked into or disposed on an outer surface of the prong or needle **304**. In at least one embodiment, the prong or needle **304** may be fluidly connected to the reservoir **302** such that the allergens may be delivered to the patient's skin **102** through the prong or needle **304**. In another embodiment, the reservoir **302** may be disposed within the body of the prong or needle **304** such that the allergens may be delivered to the patient's skin **102** by dissolution of the prong or needle after initial penetration.

(22) The at least one prong or needle **304** may extend from or extend through a first surface **306** of the reservoir **302**. The first surface **306** is disposed substantially parallel to and coplanar with the shelf **222** or surface of the capture device **112** while the first applicator **106** is in the first applicator first position, as shown in FIGS. 1 and 4.

(23) The allergens may be delivered to the first predetermined area **300** of the patient's skin **102** while the first applicator **106** is in the first applicator second position, as shown in FIG. 5. The allergens may be delivered to a patient's epidermis by the prong **304** or may be delivered intradermally by the needle **304**.

(24) In at least one embodiment, a control prong **308** may be provided with the first applicator **106** and/or the second applicator **108**. As shown in FIG. 1, the control prong **308** is provided with the first applicator **106** and is spaced apart from the prong **304**. The control prong **308** is not configured to deliver allergens but instead is arranged to prick the patient's skin **102** to provide a negative control or a negative control substance such that engagement of the control prong **308** with the first predetermined area **300** of the patient's skin **102** causes a mechanical injury to determine a patient's baseline response to a prong absent the known allergens. For example, the engagement of the control prong **308** with the first predetermined area **300** of the patient's skin **102** may lead to the formation of a small wheal that may have no or minimal clinical significance. The small wheal may serve as a baseline or a control that is compared by the allergy testing device **100** to a wheal that may form due to the delivery of allergens within the first predetermined area **300** of the patient's skin **102** by the prong **304** of the first applicator **106** and/or a wheal that may form due to the introduction of allergens that are delivered to a second predetermined area **310** of the patient's skin

102 by the second prong **314**. The comparison of the control wheal or small wheal to the wheal enables the allergy testing device **100** to determine the potential clinical significance of the wheal. (25) The second applicator **108** is spaced apart from the first applicator **106**. In the embodiments shown, the second applicator **108** is spaced apart from the first applicator **106** by the protrusion **220**. The second applicator **108** is arranged to apply antigen epitopes or the like to a second predetermined area **310** of the patient's skin **102**, as shown in FIGS. 2, 3, and 5. The second predetermined area **310** is spaced apart from the first predetermined area **300**. A footprint of or a cross sectional area of the second applicator **108** may be equal to or less than the second predetermined area **310**.

(26) The second applicator **108** includes a reservoir **312** and a prong or a needle **314**. The reservoir **312** contains allergens, histamines, antigen epitopes, saline (e.g. for negative control) or the like that will be applied to the patient skin **102** by the at least one prong or needle **314**. The at least one prong or needle **314** may be disposed within the reservoir such that the contents of the reservoir may be soaked into or disposed on an outer surface of the prong or needle **314**. In at least one embodiment, the prong or needle **314** may be fluidly connected to the reservoir **312** such that the antigen epitopes may be delivered to the patient's skin **102** through the prong or needle **314**. In another embodiment, the reservoir **312** may be disposed within the body of the prong or needle **314** such that the allergens may be delivered to the patient's skin **102** by dissolution of the prong or needle after initial penetration.

(27) The at least one prong or needle **314** may extend from or extend through a second surface **316** of the reservoir **312**. The second surface **316** is disposed substantially parallel to and coplanar with the shelf **222** or surface of the capture device **112** while the second applicator **108** is in the second applicator first position, as shown in FIGS. 1 and 4.

(28) The antigen epitopes may be delivered or applied to the second predetermined area **310** of the patient's skin **102** while the second applicator **108** is in the second applicator second position, as shown in FIG. 5. The antigen epitopes may be delivered to a patient's epidermis by the prong **314** or may be delivered intradermally by the needle **314**.

(29) The first applicator **106** and the second applicator **108** are independently moved between their respective first and second positions by the drive device **110**. The drive device **110** may be a mechanical system, an electromechanical system, a piezoelectric system, a micromechanical actuator, or a manual actuation system that is substantially disposed within the second housing member **202**.

(30) The drive device **110** includes a first guide rail **400**, a first actuator **402**, a second guide rail **404**, and a second actuator, **406**. The first guide rail **400** extends between and is connected to the first applicator **106** and the first housing member **200**. In such an embodiment, the first applicator **106** and the first guide rail **400** may be integrated members such that both may be discarded after use. In other embodiments, the first guide rail **400** extends between and is connected to the first applicator **106** and the second housing member **202**. The first guide rail **400** may be a spring assisted guide rail that is arranged to facilitate the movement of the first applicator **106** between the first applicator first position and the first applicator second position. The spring may be disposed about or provided with the first guide rail **400**. The spring is arranged to bias the first applicator **106** towards the first applicator first position.

(31) The first actuator **402** may be an electric motor or geared mechanism having a cam roller, or the like that is engaged with the first guide rail **400** or the first applicator **106**. Operation of the first actuator **402** is arranged to move the prong/needle **304** of the first applicator **106** smoothly towards and away from the patient's skin **102**.

(32) The second guide rail **404** extends between and is connected to the second applicator **108** and the first housing member **200**. In such an embodiment, the second applicator **108** and the second guide rail **404** may be integrated members such that both may be discarded after use. In other embodiments, the second guide rail **404** extends between and is connected to the second applicator

108 and the second housing member **202**. The second guide rail **404** may be a spring assisted guide rail that is arranged to facilitate the movement of the second applicator **108** between the second applicator first position and the second applicator second position. The spring be disposed about or provided with the second guide rail **404**. The spring is arranged to bias the second applicator **108** towards the second applicator first position.

(33) The second actuator **406** may be an electric motor or geared mechanism having a cam roller, or the like that is engaged with the second guide rail **404** of the second applicator **108**. Operation of the second actuator **402** moves the prong/needle **314** of the second applicator **108** smoothly towards and away from the patient's skin **102**.

(34) The capture device **112** is disposed within the protrusion **220** of the second housing member **202** of the housing **104**. The capture device **112** may extend at least partially through the shelf **222** or may be spaced apart from or extends through an opening defined within the shelf **222**.

(35) The capture device **112** includes one or more cameras **500** and a projector **502**. The camera **500** has a field of view **504** of the patient's skin **102**. The field of view **504** of the camera **500** enables the camera **500** to capture both the first predetermined area **300** and the second predetermined area **310** of the patient's skin **102**.

(36) The camera **500** may be an imaging device such as a photo detector or other detector that is arranged to capture or generate one or more images of the first predetermined area **300** and the second predetermined area **310** of the patient's skin **102**. In at least one embodiment, the camera **500** is arranged to operate in the visible light spectrum while another embodiments the camera **500** may be arranged to operate in other light spectrums such as ultraviolet, thermal, or the like.

(37) The projector **502** is arranged to project predetermined patterns onto the patient's skin **102** within at least one of the first predetermined area **300** and the second predetermined area **310**. The projector **502** includes an illumination source and a grating or pattern generator. The illumination source may emit or transmit selected wavelengths such as ultraviolet radiation, visible radiation, infrared radiation, or the like. Pattern generation may be achieved through any manner of means including, but not limited to, analogue methods, digital spatial light modulation, or laser interference.

(38) Referring to FIG. 3, the projector **502** projects a first predetermined pattern or series of patterns **510** onto the patient's skin **102** within the first predetermined area **300**. The projector **502** is also arranged to project a second predetermined pattern or series of patterns **512** onto the patient's skin **102** within the second predetermined area **310**.

(39) The set/series of patterns projected by projector **502** onto the patient's skin **102** may consist of alternating bands of at least two different electromagnetic frequencies. The frequency of band alternation may be sequentially modulated and may additionally include phase shifting and/or re-orientation of the projected pattern.

(40) Referring back to FIGS. 1, 4, and 5, the processor **114** may be provided as part of a controller that is disposed within the second housing member **202**. The processor **114** is in communication with the drive device **110**, the capture device **112**, and in some embodiments, the first applicator **106** and the second applicator **108**. The processor **114** is configured to execute program instructions maintained on a memory medium. The processor **114** may execute any of the various steps described throughout the present disclosure. Further, the processor **114** may be configured to receive data including, but not limited to, images of the first and second predetermined areas **300**, **310** of the patient's skin **102** from the camera **500** of the capture device **112**. The processor **114** may also control the indication of a positive or negative test result and may further control a display to monitor testing progress or status.

(41) The processor **114** may include a processing element such as any microprocessor-type device configured to execute algorithms and/or instructions. The term "processor" may be broadly defined to encompass any device having one or more processing elements, which execute program instructions from a non-transitory memory medium. Further, the steps described throughout the

present disclosure may be carried out by a single processor/controller or multiple processors/controllers.

(42) The memory medium may include any storage medium known in the art suitable for storing program instructions executable by the processor **114**. For example, the memory medium may include a non-transitory memory medium or non-transitory computer readable medium. By way of another example, the memory medium may include, but is not limited to, read-only memory, random access memory, magnetic or optical memory device, magnetic tape, solid state drive, or the like. The memory medium may be housed together with the processor **114**. In at least one embodiment, the memory medium may be located remotely with respect to the processor **114**. For instance, the processor **114** may access remote memory such as a server that is accessible through a network.

(43) The processor **114** may perform the steps discussed below and shown in conjunction with the flow chart of FIG. **6**. The processor **114** may: operate the projector **502** of the capture device **110** to project a first predetermined set/series of patterns **510** onto the patient's skin **102** within the first predetermined area **300** and/or the second predetermined area **310**; operate the camera **500** of the capture device **110** to capture a first image set/series of the first predetermined area **300** and/or the second predetermined area **310** of the patient's skin **102**; operate the drive device **110** to move the first applicator **106** from the first applicator first position towards the first applicator second position to apply allergens to the patient's skin **102** within the first predetermined area **300**; after a predetermined period of time, operate the projector **502** of the capture device **110** to project a second predetermined set/series of patterns **512** onto the patient's skin **102** within the first predetermined area **300**; operate the camera **500** of the capture device **110** to capture a second image set/series of the first predetermined area **300** of the patient's skin **102**; determine the significance of the second image set/series; operate the drive device **110** to move the second applicator **108** from the second applicator first position towards the second applicator second position to apply allergens or antigen epitopes to the patient's skin **102** within the second predetermined area **310**; after a second predetermined period of time, operate the projector **502** of the capture device **110** to project a third predetermined pattern set/series **514** onto the patient's skin **102** within the second predetermined area **310**; operate the camera **500** of the capture device **110** to capture a third image set/series of the second predetermined area **310** of the patient's skin **102**; determine the clinical significance of the second image set/series; and operate an indicator **520** to output for display whether an allergic reaction has occurred within at least one of the first predetermined area **300** and/or the second predetermined area **310**.

(44) FIG. **6** is a flow chart illustrating steps performed in an allergy testing method **600**, in accordance with one or more embodiments of the present disclosure. Applicant notes that the embodiments and enabling technologies described previously herein in the context of the allergy testing device **100** should be interpreted to extend to the method **600**. It is further noted, however, that the method **600** is not limited to the architecture of the allergy testing device **100**.

(45) The method includes positioning the allergy testing device **100** on a body part of a patient, such as the forearm of a patient, at step **602**. The allergy testing device **100** is positioned such that the capture device has a clear and unobstructed view of the patient's skin **102**. In some embodiments, the method includes an initializing step **604** in which the capture device **112** determines or identifies the first predetermined area **300** and the second predetermined area **310** on the patient's skin. Identifying the first predetermined area **300** and the second predetermined area **310** may include calculating an initial map in cartesian coordinates, a grid, or the like. The initial map or grid will aid the processor to correlate a prong/needle of the first or second applicator **106**, **108** to a specific location or region after application of the allergen/antigen epitopes to a patient's skin **102**.

(46) The method further includes a first capture step **606** that includes projecting the first series/set of patterns **510** on the patient's skin **102** using the projector **502** and capturing a first set/series of

images of the first predetermined area **300** using the camera **500**. The first capture step **606** provides an initial or baseline capture of the first predetermined area **300** and in some embodiments may also include capturing a first set/series of images of the second predetermined area **310** of the patient's skin using the projector **502** and camera **500** of the capture device **112**.

(47) The method further includes a first application step **608** that includes applying an allergen to the patient's skin **102** within the first predetermined area **300** of the patient's skin **102** with the first applicator **106**. The allergen may be applied to the patient's epidermis by a prong or applied intradermally by a needle.

(48) The method further includes a second capture step **610** that includes projecting the second series/set of patterns **512** and capturing a second set/series of images of the first predetermined area **300** of the patient's skin **102** using the projector **502** and camera **500** of the capture device **112** after a first predetermined time period. The first predetermined time period provides a sufficient amount of time for a potential allergic reaction to occur such that a wheal may begin formation within the first predetermined area **300** of the patient's skin **102**.

(49) The method further includes a first determination step **612** that may include comparing the first image set/series and the second image set/series to each other. In at least one embodiment, should an allergic reaction occur such that a wheal forms within the first predetermined area **300**, the method may assign coordinates, a grid location, or the like to the wheal.

(50) The determination step **612** may be a multistep process that involves first transforming each image set/series from the first capture step **606** and/or the second capture step **610** into a discrete, interpreted object. Transformation of a capture set/series is achieved by signal analysis of the projected and captured patterns visible on the patient's skin **102**. The captured pattern in each image of the set/series from the first capture step **606** and/or the second capture step **610**, and knowledge of the expected pattern frequency projected in association with each image, are used to calculate the location, size, and features of skin topography of the patient's skin **102** within the first predetermined area **300** which created distortions from the projected pattern's expected appearance. Information about distortions in each image of the set/series is aggregated to generate a three-dimensional point cloud representing the surface of the patient's skin **102**. In this way, the entire set/series of the first capture step **606** and/or the second capture step **610** is transformed into a unified point cloud. A series of data cleaning and reconstitution steps are undertaken to optimize the data including correction of natural noise, skin inter-reflections, and subsurface scattering. Anomaly detection is then processed on the generated point cloud to identify changes in skin topography including allergic wheals as well as benign appearances such as nevi, skin tags, and the like. Features of anomalous topography, including size and location, are recorded. These features are further used to perform parametric fitting of the anomalous topography to create interpretations containing objects of interest. If appropriate, the processor **114** compares post-exposure interpretations (those generated from capture set/series collected after operation of the applicators), to baseline interpretations (those generated from capture set/series collected before operation of the applicators) and/or interpretations collected previous to the most recent application step to identify those objects of interest which represent benign skin topography such as nevi, skin tags, and the like or wheals developed in response to previous applications. When indicated, the processor **114** then uses prior knowledge of control substance application position to compare intra-interpretation objects of interest resulting from control substance application to the remaining objects of interest identified as neither benign nor control objects. Finally, the device uses feature information derived from these non-benign, non-control objects, in addition to thresholding information, to determine the clinical significance of the interpretation through feature analysis compared against the threshold.

(51) As understood by one of ordinary skill in the art, clinical significance means that an identifiable wheal has formed or has begun forming such that the skin topography may have changed due an allergic reaction. Clinical significance may be differentiated from non-allergic

changes by feature analysis of the identified change compared to the current topography of the patient's skin **102** at the site of negative control substance exposure, e.g. control prong **308**. A wheal is considered clinically significant for an allergic sensitivity when it is larger than a wheal induced by the negative control substance by a predetermined threshold. Alternatively, if the negative control substance does not induce a wheal, or if a negative control substance is not applied as part of the test, then a wheal is considered clinically significant for allergic sensitivity when it is simply larger than a predetermined threshold. Current literature and professional guidance recommend a threshold of **3** millimeters.

(52) Should the determination derived from the image set/series be considered significant or to have clinical significance at step **612**, the method continues to the first output step **614**. The first output step **614** includes outputting for display a first indicator indicative of an allergic reaction via the indicator **520**. The first indicator may be an auditory, visual, or other indicator that may be output by the allergy testing device **100**. The method may then end If a difference between the first image set/series and the second image set/series is less than the threshold, the method continues to the second application step **616**.

(53) The second application step **616** includes applying or delivering allergens or antigen epitopes within the second predetermined area **310** of the patient's skin **102** with the second applicator **108**. The allergens or antigen epitopes may be applied to the patient's epidermis by a prong or applied intradermally by a needle.

(54) The method further includes a third capture step **618** that includes projecting the third series/set of patterns **514** on the patient's skin **102** using the projector **502** and capturing a third set/series of images of the second predetermined area **310** using the camera **500** after a second predetermined time period. The second predetermined time period provides a sufficient amount of time for a potential allergic reaction to occur such that a wheal may be formed within the second predetermined area **310** of the patient's skin **102**. The second predetermined time period may be substantially equal to the first predetermined time period or may be different from the first predetermined time period.

(55) The method further includes a second determination step **620** that may include comparing the third image set/series to at least one of the first image set/series and the second image set/series. In at least one embodiment, should an allergic reaction occur such that a wheal forms within the second predetermined area **310**, the method may assign coordinates, grid location, or the like to the wheal.

(56) Should the determination derived from the third image set/series be considered significant, the method continues to the second output step **622**. The second output step **622** includes outputting for display a second indicator indicative of an allergic reaction via the indicator **520**. The second indicator may be an auditory, visual, or other indicator that may be output by the allergy testing device **100**. If a difference between the third image set/series to at least one of the first image set/series and the second image set/series be less than the threshold, the method may output for display a third indicator indicative of no allergic reaction at **624** via the indicator **520**.

(57) One skilled in the art will recognize that the herein described components (e.g., operations), devices, objects, and the discussion accompanying them are used as examples for the sake of conceptual clarity and that various configuration modifications are contemplated. Consequently, as used herein, the specific exemplars set forth and the accompanying discussion are intended to be representative of their more general classes. In general, use of any specific exemplar is intended to be representative of its class, and the non-inclusion of specific components (e.g., operations), devices, and objects should not be taken as limiting.

(58) Those having skill in the art will appreciate that there are various vehicles by which processes and/or systems and/or other technologies described herein can be effected (e.g., hardware, software, and/or firmware), and that the preferred vehicle will vary with the context in which the processes and/or systems and/or other technologies are deployed. For example, if an implementer

determines that speed and accuracy are paramount, the implementer may opt for a mainly hardware and/or firmware vehicle; alternatively, if flexibility is paramount, the implementer may opt for a mainly software implementation; or, yet again alternatively, the implementer may opt for some combination of hardware, software, and/or firmware. Hence, there are several possible vehicles by which the processes and/or devices and/or other technologies described herein may be effected, none of which is inherently superior to the other in that any vehicle to be utilized is a choice dependent upon the context in which the vehicle will be deployed and the specific concerns (e.g., speed, flexibility, or predictability) of the implementer, any of which may vary.

(59) The previous description is presented to enable one of ordinary skill in the art to make and use the present disclosure as provided in the context of a particular application and its requirements. As used herein, directional terms such as “left”, “right”, “top,” “bottom,” “over,” “under,” “upper,” “upward,” “lower,” “down,” and “downward” are intended to provide relative positions for purposes of description and are not intended to designate an absolute frame of reference. Various modifications to the described embodiments will be apparent to those with skill in the art, and the general principles defined herein may be applied to other embodiments. The present disclosure is not intended to be limited to the particular embodiments shown and described but is to be accorded the widest scope consistent with the principles and novel features herein disclosed.

(60) With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations are not expressly set forth herein for sake of clarity.

(61) All of the methods described herein may include storing results of one or more steps of the method embodiments in memory. The results may include any of the results described herein and may be stored in any manner known in the art. The memory may include any memory described herein or any other suitable storage medium known in the art. After the results have been stored, the results can be accessed in the memory and used by any of the method or system embodiments described herein, formatted for display to a user, used by another software module, method, or system, and the like. Furthermore, the results may be stored “permanently,” “semi-permanently,” temporarily,” or for some period of time. For example, the memory may be random access memory (RAM), and the results may not necessarily persist indefinitely in the memory.

(62) It is further contemplated that each of the embodiments of the method described above may include any other step(s) of any other method(s) described herein and may also be performed in a different order. In addition, each of the embodiments of the method described above may be performed by any of the systems described herein.

(63) The herein described subject matter sometimes illustrates different components contained within, or connected with, other components. It is to be understood that such depicted architectures are merely exemplary, and that in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “connected,” or “coupled,” to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “couplable,” to each other to achieve the desired functionality. Specific examples of couplable include but are not limited to physically able to mate and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interacting and/or logically interactable components.

(64) Furthermore, it is to be understood that the present disclosure is defined by the appended claims. It will be understood by those within the art that, in general, terms used herein, and

especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” and the like). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to inventions containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, and the like” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, and the like). In those instances where a convention analogous to “at least one of A, B, or C, and the like” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, and the like). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

(65) It is believed that the present disclosure and many of its attendant advantages will be understood by the foregoing description, and it will be apparent that various changes may be made in the form, construction and arrangement of the components without departing from the disclosed subject matter or without sacrificing all of its material advantages. The form described is merely explanatory, and it is the intention of the following claims to encompass and include such changes.

Claims

1. An allergy testing device, comprising: a housing; a first applicator configured to apply at least one allergen to a first predetermined area of the patient's skin, the first applicator being operatively connected to the housing and movable between a first applicator first position and a first applicator second position; a second applicator configured to apply the at least one allergen to a second predetermined area of the patient's skin, the second applicator being operatively connected to the housing and spaced apart from the first applicator; a drive device operatively connected to each of the first applicator and the second applicator, the drive device configured to move the first applicator to the first predetermined area and to move the second applicator to the second predetermined area, wherein the drive device is further configured to move the first applicator between the first applicator first position and the first applicator second position; a capture device

operatively connected to the housing and spaced apart from the first applicator and the second applicator; and at least one processor disposed within the housing and in communication with the drive device and the capture device, wherein the processor comprises a non-transitory computer readable medium that stores computer executable instructions, such that the processor is configured to execute the computer executable instructions embodied in the non-transitory computer readable medium.

2. The allergy testing device of claim 1, wherein the second applicator is movable between a second applicator first position and a second applicator second position; and wherein the drive device is configured to move the second applicator between the second applicator first position and a second applicator second position.

3. The allergy testing device of claim 2, wherein: the first applicator includes at least one prong or needle arranged to apply the at least one allergen to a patient's skin while in the first applicator second position; and the second applicator includes at least one prong or needle arranged to apply the at least one allergen to a patient's skin while in the second applicator second position.

4. The allergy testing device of claim 3, wherein the processor is further programmed to: operate the capture device to project a first predetermined set of patterns onto the patient's skin within the first predetermined area and the second predetermined area; operate the capture device to capture a first set of images of the first predetermined area and the second predetermined area of a patient's skin; and operate the drive device to move the first applicator from a first applicator first position towards a first applicator second position, within the first predetermined area.

5. The allergy testing device of claim 4, wherein the processor is further programmed to: operate the capture device to project a second predetermined set of patterns onto the patient's skin within the first predetermined area; operate the capture device to capture a second set of images of the first predetermined area; and evaluate the second set of images and render a first determination, wherein the first determination is clinically significant when at least one wheal larger than a predetermined threshold size is detected in the second set of images and wherein the first determination is clinically insignificant when each wheal detected in the second set of images is below the predetermined threshold size.

6. The allergy testing device of claim 5, wherein the processor is further programmed to: responsive to the first determination derived from the second image set being considered clinically insignificant, output for display a first indicator, indicative of no allergic response.

7. The allergy testing device of claim 6, wherein the processor is further programmed to: operate the drive device to move the second applicator from the second applicator first position towards the second applicator second position, within the second predetermined area in response to the determination derived from the second image set being considered clinically insignificant.

8. The allergy testing device of claim 7, wherein the processor is further programmed to: operate the capture device to project a third predetermined pattern set on the patient's skin within the second predetermined area; operate the capture device to capture a third image set of the second predetermined area of the patient's skin; and evaluate the third image set and render a second determination, wherein the second determination is clinically significant when at least one wheal larger than a predetermined threshold size is detected in the third set of images and wherein the second determination is clinically insignificant when each wheal detected in the third set of images is below the predetermined threshold size.

9. The allergy testing device of claim 8, wherein the processor is further programmed to: responsive to the second determination derived from the third image set being considered clinically significant, output for display a second indicator, indicative of an allergic response.

10. The allergy testing device of claim 8, wherein the processor is further programmed to: responsive to the second determination derived from the third image set being considered clinically insignificant, output for display a third indicator, indicative of no allergic response.

11. The allergy testing device of claim 6, wherein the first indicator is at least one of an auditory

indicator and a visual indicator.

12. The allergy testing device of claim 9, wherein the second indicator is at least one of an auditory indicator and a visual indicator.

13. An allergy testing device, comprising: a housing; a first applicator movable between a first applicator first position and a first applicator second position and configured to apply at least one allergen to a first predetermined area of the patient's skin, the first applicator being operatively connected to the housing; a drive device operatively connected to the first applicator and configured to move the first applicator between the first applicator first position and the first applicator second position with the first predetermined area; a capture device operatively connected to the housing and spaced apart from the first applicator; and at least one processor disposed within the housing and in communication with the drive device and the capture device, wherein the processor comprises a non-transitory computer readable medium that stores computer executable instructions, such that the processor is configured to execute the computer executable instructions embodied in the non-transitory computer readable medium, wherein the processor is further programmed to: operate the capture device to project a first predetermined set of patterns onto the patient's skin within the first predetermined area; operate the capture device to capture a first set of images of the first predetermined area; operate the drive device to move the first applicator from a first applicator first position towards a first applicator second position, within the first predetermined area; operate the capture device to project a second predetermined set of patterns onto the patient's skin within the first predetermined area; operate the capture device to capture a second set of images of the first predetermined area; and evaluate the second set of images and render a first determination, wherein the first determination is clinically significant when at least one wheal larger than a predetermined threshold size is detected in the second set of images and wherein the first determination is clinically insignificant when each wheal detected in the second set of images is below the predetermined threshold size.

14. The allergy testing device of claim 13 wherein the first applicator is a prong.

15. The allergy testing device of claim 13 wherein the first applicator is a needle.

16. The allergy testing device of claim 13, wherein the processor is further programmed to: responsive to the first determination derived from the second image set being considered clinically significant, output for display a first indicator, indicative of an allergic response.

17. The allergy testing device of claim 13, wherein the processor is further programmed to: responsive to the first determination derived from the second image set being considered clinically insignificant, output for display a first indicator, indicative of no allergic response.

18. The allergy testing device of claim 17, further comprising: a second applicator movable between a second applicator first position and a second applicator second position and configured to apply the at least one allergen to a second predetermined area of the patient's skin, the second applicator operatively connected to the housing and spaced apart from the first applicator; and wherein the drive device is operatively connected to the second applicator and arranged to move the second applicator between the second applicator first position and a second applicator second position within the second predetermined area.

19. The allergy testing device of claim 18 wherein the second applicator is a prong.

20. The allergy testing device of claim 18 wherein the second applicator is a needle.

21. The allergy testing device of claim 18, wherein the processor is further programmed to: operate the drive device to move the second applicator from the second applicator first position towards the second applicator second position, within the second predetermined area in response to the determination derived from the second image set being considered clinically insignificant; operate the capture device to project a third predetermined pattern set on the patient's skin within the second predetermined area; operate the capture device to capture a third image set of the second predetermined area of the patient's skin; and evaluate the third image set and render a second determination, wherein the second determination is clinically significant when at least one wheal

larger than a predetermined threshold size is detected in the third set of images and wherein the second determination is clinically insignificant when each wheal detected in the third set of images is below the predetermined threshold size.

22. The allergy testing device of claim 21, wherein the processor is further programmed to: responsive to the second determination derived from the third image set being considered clinically significant, output for display a second indicator, indicative of an allergic response.

23. The allergy testing device of claim 21, wherein the processor is further programmed to: responsive to the second determination derived from the third image set being considered clinically insignificant, output for display a third indicator, indicative of no allergic response.
